

Nederman Dust Collection System Equipment & Nameplate Documentation Package

Unique Wood Products

Prepared for ducting / CFM engineering and control panel design

Facility	Unique Wood Products (UWP)
System	Nederman dust collection system (filter unit, fan section, rotary valves, screw conveyor / transfer drives)
Prepared for	Carl — Air pollution control (duct design, CFM / airflow sizing, facility tie-in) Canfil — Control panel design and motor controls
Prepared by	Unique Wood Products — mylifepast30@gmail.com
Date	August 7, 2026
Status	PARTIAL — 5 motors on top of the collector (fan / filter section) are not yet documented. See Section 8.

This package contains every motor and component nameplate we have been able to photograph and transcribe to date, plus a consolidated electrical schedule and a list of the open items we still need. Please review Section 2 (what we need from you) first.

1. Purpose of This Package

Unique Wood Products is installing / re-commissioning a Nederman dust collection system and needs engineering support in two areas. This document is the single reference for both scopes, so the same equipment data is in front of everyone.

1.1 Scope for Carl — Air Pollution / Airflow Engineering

- Duct design and routing from the woodworking machinery inside the Unique Wood Products building out to the Nederman collector (collector sits outside; see photos).
- CFM requirements per drop / per machine, total system CFM, and confirmation the Nederman filter unit and fan(s) are sized for it.
- Static pressure calculations for the duct runs, elbows, and filter media so fan performance can be verified.
- Transport velocity check for wood dust (typically 3,500–4,500 FPM in branch and main ducts) to avoid material settling.
- How and where the ductwork ties into the building when we come back — wall / roof penetrations, drop locations, blast gate locations.
- Review of explosion protection requirements for the duct system (isolation valves, backdraft dampers, venting) given the St2-rated wood dust — NFPA 652 / NFPA 664 apply.

1.2 Scope for Canfil — Control Panel

- Design and build of the main control panel for the complete dust collection system.
- Motor starters / VFDs for every motor listed in the electrical schedule (Section 3) — including the 5 top-mounted motors still to be documented.
- Note the Hoyer auxiliary motor is a 50 Hz nameplate (230/400 V) — it will need a VFD or a confirmed supply arrangement on our 460 V / 60 Hz service (Section 4).
- Start/stop sequencing and interlocks: main fan(s), rotary valves, screw conveyor / transfer drives, filter cleaning system.
- Differential pressure monitoring across the filter, level indication on the discharge, and ATEX / hazardous-location wiring where required (rotary valve is ATEX-marked, dust class St2).
- Disconnects, overloads, and control voltage per the final motor schedule.

Bottom line: Carl figures out the ducting and CFMs and how we tie into the building; Canfil builds the panel that runs all of the motors. Both need the motor data in this package, and both should flag anything missing that they need from Nederman.

2. Equipment Inventory — What Is Documented

The system is an outdoor Nederman dust collector (blue Nederman-branded housing) with galvanized ducting, at least one rotary discharge valve, a screw conveyor / transfer arrangement, and multiple motors. The table below lists every unit covered in this package and its status.

#	Component	Manufacturer / Type	Status
1	Auxiliary drive motor (mounted under duct)	Hoyer HMA3 90L-6 — 1.1 kW	Documented — Section 4
2	Rotary valve / conveyor gearmotor	JIE JDRIVE DL80M4/P/D160 — 1 HP with inline helical reducer	Documented — Section 5
3	Rotary discharge valve	Nederman NRSZ-10 ATEX / NRSZ 10-0	Documented — Section 6
4–8	Five (5) motors on top of the collector — fan / filter-cleaning section	Unknown — could not access (roof/lift required)	NOT YET DOCUMENTED — Section 8



Overall view: JIE gearmotor (center, gray) driving the Nederman rotary valve on the blue housing; black Hoyer-type motor visible top of frame; galvanized transfer piping below.

3. Consolidated Motor / Electrical Schedule (for Canfil)

All motor electrical data known to date. Site service is assumed 460 V / 3-phase / 60 Hz (the Nederman rotary valve plate is a 460/3/60 unit). Canfil: please confirm service details with us before final panel design.

Motor	HP / kW	Voltage	Hz	FLA	RPM	Encl.	Notes
Hoyer HMA3 90L-6	1.5 HP / 1.1 kW	230/400 V Δ/Y	50	5.00 / 2.85 A	960	IP55	VFD — the one (1) VFD in the panel. Per Carl, may not be strictly necessary (see Section 4 note); confirm before ordering
JIE DL80M4/P/D160 gearmotor	1 HP / 0.75 kW	230/460 V YY/Y	60	2.9 / 1.4 A	1720	IP56	Across-the-line full-voltage starter — NO VFD (UWP decision). Flange-mount on inline reducer; drives rotary valve/conveyor
Nederman NRSZ-10 rotary valve drive	1 HP	460 V	60	TBD	20 valve rpm	ATEX	ATEX Ex II 1/-D, dust St2, T125 °C; cert FTZU 12 ATEX 0142X
Top motor #1—#5 (fan / filter section)	TBD	TBD	TBD	TBD	TBD	TBD	Nameplates not yet accessible — data to follow (Section 8)

Control / safety items for panel design

- Suggested run sequence: rotary valve + transfer drives start first, then main fan; fan proves airflow (DP or current) before machines release.
- Filter differential-pressure gauge/transmitter input for cleaning-cycle control and filter-loading alarm.
- Rotary valve is the explosion isolation device on the discharge — it must run whenever the fan runs; interlock accordingly.
- Dust is St2 class (wood dust). Panel and field wiring in/near the dirty-air side may need hazardous-location ratings — Canfil to advise per NFPA 652/664 and the ATEX markings shown in Section 6.
- **VFD decision (UWP, 2026-08-07):** panel gets ONE VFD, on the smaller Hoyer motor only. The JIE gearmotor runs across-the-line at full voltage. Carl has indicated even the one VFD may not be necessary — Canfil please confirm and quote both ways.
- Starter type for the 5 top motors to be decided once their nameplates are captured; if any are also 50 Hz-only European plates, revisit with Carl.

4. Motor 1 — Hoyer HMA3 90L-6 (Auxiliary Drive)

Foot-mounted 3-phase motor installed horizontally under the galvanized duct/conveyor housing (photo next page). Full nameplate transcription:

Manufacturer	Hoyer Motors — www.hoyermotors.com
Type	HMA3 90L-6 (3-phase induction motor, IE3 premium efficiency)
Serial number	DK1040430237
Rated power	1.1 kW ($\approx$ 1.5 HP)
Speed	960 rpm (6-pole at 50 Hz)
Voltage	230 V Δ / 400 V Y
Frequency	50 Hz (no 60 Hz rating shown on plate)
Full-load current	5.00 A (Δ) / 2.85 A (Y)
Power factor ($\cos \phi$)	0.68
Efficiency (IE3)	81.0% @ 4/4 load, 80.3% @ 3/4, 76.8% @ 1/2 — per IEC 60034-1
Duty	S1 (continuous), 40 °C ambient
Enclosure	IP55
Insulation	Class F (temperature rise class B)
Mounting	IM B3 (foot-mounted)
Bearings	Drive end 6205, non-drive end 6205
Weight	20.4 kg
Manufacture date	Week 47, 2022 (YY/MW 22/47)

VFD decision (UWP): this is the one motor in the panel that gets a VFD; everything else starts across-the-line at full voltage. Note from Carl: the VFD may not be strictly necessary — a 400 V / 50 Hz motor on a 460 V / 60 Hz service keeps essentially the same volts-per-hertz ratio ($400/50 = 8.0$ vs $460/60 \approx 7.7$), so it can run direct-on-line at $\sim 1,150$ rpm (20% faster than nameplate) with rated torque. Whether that speed is acceptable depends on what this motor drives — Canfil / Nederman to confirm before the VFD is dropped. Until confirmed, carry the VFD in the design.

4.1 Hoyer Motor — Nameplate Photo



Hoyer HMA3 90L-6 nameplate (image rotated 180° for readability).

4.2 Hoyer Motor — As Installed



Hoyer motor mounted horizontally beneath the galvanized housing on angle brackets; conduit exits the terminal box to the right. Blue Nederman equipment visible below.

5. Motor 2 — JIE JDRIVE DL80M4/P/D160 Gearmotor

Flange-mounted (IM B5) motor on an inline JIE helical gear reducer, driving the Nederman rotary valve / transfer screw (see overview photo, Section 2). Full nameplate transcription:

Manufacturer	JIE USA Inc. — www.jie-drives.com — +1 630-580-9986
Description	Three-phase induction motor, IEC 60034
Type	DL80M4/P/D160
Unit number	100132971-00023
Rated power	1 HP / 0.75 kW
Speed	1720 rpm (4-pole at 60 Hz)
Voltage	230 V (YY) / 460 V (Y)
Frequency	60 Hz
Full-load current	2.9 A / 1.4 A (230/460 V)
Power factor (cos ϕ)	0.79
Efficiency (η)	82.5%
Duty	CONT (continuous, S1)
Enclosure	IP56
Insulation	Class F
Mounting	IM B5 (flange)
Weight	13.8 kg / 30 lb
Production date (POD)	2021-05
Gear reducer	JIE inline helical unit (casting marked JD44xxxx); output to rotary valve shaft

This motor is dual-rated 230/460 V at 60 Hz. **UWP decision: it runs across-the-line at full voltage with a standard starter — no VFD on this drive.** FLA at 460 V is 1.4 A.

5.1 JIE Gearmotor — Nameplate Photo



JIE JDRIVE nameplate on the gearmotor fin housing. QR code links to unit record; phone number on plate is JIE USA support.

6. Nederman Rotary Discharge Valve (ATEX)

The rotary valve on the collector discharge carries two Nederman labels side by side — a US label (Nederman Manufacturing & Logistics LLC, Charlotte, NC) and an EU label (Nederman Manufacturing Poland Sp. z o.o.). Both are transcribed below. This valve is an ATEX-certified explosion isolation component — relevant to both the duct design and the control panel.

6.1 US label — NRSZ-10 ATEX/460V

Name / Type	Rotary Valve — NRSZ-10 Atex/460v
Part number	89117006
Serial / prod. year	22650601 — 2024
Weight	180 kg / 396 lb
Power	1 HP
Power supply	460 V / 3-phase / 60 Hz
Valve speed (max)	20 rpm
T max	125 °C / 257 °F
Pred. max	25 kPa (St1) / 20 kPa (St2)
Dust type	St2
ATEX marking	Ex II 1/- D C St2 T125
Certificate	FTZU 12 ATEX 0142X — CE 2503
Manufacturer	Nederman Manufacturing and Logistics LLC, 4500 Chesapeake Drive, Charlotte, NC 28216 — (800) 533-5286

6.2 EU label — NRSZ 10-0

Name / Type	Rotary valve — NRSZ 10-0
Part number	73008427
Serial / prod. year	19965706 — 2023
Weight	95 kg
Valve rpm / rpm max	20 / 50
Max capacity	3.5 (m ³ /h per rpm row on plate — see photo)
Pred. max	(St1) 25 kPa / (St2) 20 kPa
ATEX marking	D St2 — certificate FTZU 12 ATEX 0142X — CE 1026
Manufacturer	Nederman Manufacturing Poland Sp. z o.o., PL 05-270 Marki, ul. Okólna 45A — tel. +48 22 76 16 000

6.3 Rotary Valve — Label Photo



Both Nederman rotary valve labels on the blue valve body (US label left, EU label right).

7. What We Could NOT Document Yet — 5 Motors on Top

There are five (5) additional motors on top of the dust collector — the main fan(s) and the filter-cleaning section. They could not be reached safely for nameplate photos (lift / roof access required). These are almost certainly the largest loads in the system, so the panel sizing and the fan CFM verification both depend on them.

Until those plates are captured, please work from the following and flag what you need:

- **Carl:** Nederman can confirm the fan model, design CFM, and static pressure curve from the unit serial number. We will also send you the Nederman order paperwork we received so you have the as-sold configuration.
- **Canfil:** assume the 5 unknown motors are 3-phase 460 V loads and reserve panel space accordingly; final starter/VFD sizing to follow when we get the nameplates.
- **UWP (us):** get lift access, photograph all 5 nameplates plus the main Nederman unit nameplate on the housing, and forward the supplement to both of you.

Open questions for Carl

- Total system CFM required for the UWP machine lineup, and per-drop CFMs.
- Duct sizes, routing, and transport velocities from the building to the collector.
- Static pressure budget and confirmation the Nederman fan(s) can deliver it.
- Where and how we penetrate the building when we tie in on the return visit.
- Explosion isolation / venting requirements on the duct runs (St2 wood dust).

Open questions for Canfil

- Panel main size / bus rating given the known loads plus five TBD motors.
- Confirm the one-VFD plan: VFD on the Hoyer motor only, all other drives across-the-line (Carl suggests the VFD may be droppable — please verify and quote both ways).
- Interlock and sequencing scheme (Section 3) — please review and comment.
- Hazardous-location requirements for field devices near the St2 dust zones.

8. Reference & Contacts

Nederman US	Nederman Manufacturing and Logistics LLC, 4500 Chesapeake Drive, Charlotte, NC 28216 — (800) 533-5286 — www.nederman.com
Nederman Poland	Nederman Manufacturing Poland Sp. z o.o., PL 05-270 Marki, ul. Okólna 45A — +48 22 76 16 000
Hoyer Motors	www.hoyermotors.com (motor serial DK1040430237)
JIE USA Inc.	www.jie-drives.com — +1 630-580-9986 (unit no. 100132971-00023)
ATEX certificate	FTZU 12 ATEX 0142X (rotary valve)
UWP contact	mylifepast30@gmail.com

Attachments / also being sent separately

- Original full-resolution nameplate photos (5 files, IMG_6281–IMG_6285).
- Nederman order / quote paperwork as received by UWP (to follow in the same email).
- Supplement with the 5 top-motor nameplates and main unit nameplate (to follow after lift access).

Document prepared 2026-08-07 from field photos taken at the collector. All values transcribed directly from nameplates; where a value was hard to read on the photo it is noted in-line. Please verify critical electrical values against the physical plates before final design.